

MILTECH / Techaya Switch — DHCP Troubleshooting Field Guide

Offline reference — ping works, DHCP doesn't

0. IMMEDIATE FALLBACK — Get devices talking right now, no DHCP needed

Since ping already works, IP routing between your devices is fine. You can bypass DHCP entirely by assigning static IPs manually.

Step 1 — Find a valid address range. On a device that *can* currently ping others, run one of:

- Windows: `ipconfig /all` and `route print`
- Linux/Mac: `ip addr` and `ip route` (or `route -n`)

Note the subnet (e.g. 192.168.105.x /24) and the default gateway address.

Step 2 — Assign static IPs to the devices that fail DHCP. Pick an unused address in the same subnet as the gateway you found, and set the same subnet mask and gateway.

- **Windows:** Control Panel → Network → Adapter Properties → IPv4 → "Use the following IP address." Enter IP, subnet mask, gateway. DNS can be left blank or set to the gateway.

- **Windows (CLI, faster):**

```
netsh interface ip set address "Ethernet" static 192.168.105.50 255.255.255.0 192.168.105.1
```

- **Linux:**

```
sudo ip addr add 192.168.105.50/24 dev eth0
```

```
sudo ip route add default via 192.168.105.1
```

Give every device that needs to talk to each other an address in the **same** subnet/VLAN if possible. This sidesteps the DHCP/VLAN issue completely and is the most reliable option if you're short on time.

1. Identify your exact model

Check the chassis label/engraving. The MILTECH family varies significantly by model:

Model	Type	Notes
MILTECH 304 / 308	Basic switch, 8x 10/100	Limited/no IP management on some variants
MILTECH 908	8-port managed L2	DHCP relay/option 82 is an optional feature, only on part number 1-908-101
MILTECH 9012 / 9012C / 9012C/G	12-port managed L2/L3 switch + internal SBC running Linux + Cisco IOS router	Full router-on-a-stick topology, see below
MILTECH 9022	12x copper + 4x fiber, L3 managed	Static/RIPv1/v2/OSPF routing
MILTECH 9028	12-port, IP67	Managed L2/L3

If you can't find a label, the **default management IP 192.168.1.111 (mask 255.255.255.0)** is common across most managed variants — try that first.

2. Default management access

Access method	Setting
Web / Telnet	192.168.1.111 / 255.255.255.0 (switch management)
Serial (switch side)	Connect to physical Port #1, RS-232, 115200 baud, 8 data bits, No parity, 1 stop bit, No flow control
Switch login	Username: <code>admin</code> , Password: <i>(blank — just press Enter)</i>
Secondary switch IP (9012C/G only, for SBC reachability)	192.168.101.254 / 255.255.255.0
SBC (Linux) login (9012C/G only)	Username: <code>root</code> , Password: 123456
Cisco router CLI access (9012C/G only)	From Linux prompt type <code>cisco</code> , press Enter, then <code>enable</code> , password <code>cisco</code>

If `admin` with a blank password doesn't work, the unit may have been previously configured with a custom password — in that case the static-IP fallback in Section 0 is your best bet without a factory reset.

3. Root cause: why ping works but DHCP doesn't (VLAN-per-port design)

On the managed 9012-family units, **each physical port ships on its own separate VLAN by default:**

Port 1 = VLAN 1 (gateway 192.168.101.1)

Port 2 = VLAN 2 (gateway 192.168.102.1)

Port 3 = VLAN 3 (gateway 192.168.103.1)

...

Port 11 = VLAN 11 (gateway 192.168.111.1)

Port 12 = Trunk to internal router/SBC (carries all VLANs)

An internal router (Cisco IOS on the SBC) routes **unicast** traffic between these VLANs — which is why ping across ports/VLANs works. But DHCP relies on **broadcast** traffic, which does not cross VLAN boundaries by itself. If your DHCP server is on a different port/VLAN than the client that's failing, the DHCP broadcast simply never reaches it — even though the router would happily route a ping between them.

Fix, if you have management access: configure a DHCP relay (Cisco term: `ip helper-address`) on the router subinterface for the client's VLAN, pointing at the DHCP server's real IP address:

`interface Ethernet0/0.3` ! example: VLAN 3 (Port 3)

`ip helper-address 192.168.105.10` ! real IP of your DHCP server

Do this for every VLAN/port that needs DHCP from a server sitting on a different port.

Fix, if you don't have time/access to configure this: physically move the DHCP server and the affected client onto the same port group/VLAN, or just use the static-IP fallback in Section 0.

4. Show / verification commands

A. Cisco IOS router side (9012C/G — access via `cisco` then `enable` as in Section 2)

These are standard Cisco IOS commands, confirmed applicable since the embedded router runs Cisco IOS 15.6(3)M1 (5921 ESR):

Command	What it tells you
<code>show ip interface brief</code>	Lists all VLAN subinterfaces (Eth0/0.1–0/0.11), their IPs, and up/down status
<code>show running-config</code>	Full config — look for <code>ip helper-address</code> lines under each subinterface
<code>show ip interface Ethernet0/0.X</code>	For one specific VLAN subinterface — explicitly shows "Helper address is ..." if a relay is configured, plus IP/mask
<code>show ip route</code>	Confirms the router has a route to every VLAN subnet (101.0–111.0/24)
<code>show arp</code>	Confirms whether the router has learned the MAC/IP of devices on each VLAN — if a device never appears here, its broadcasts/traffic aren't reaching the router at all
<code>show interfaces Ethernet0/0</code>	Status/error counters on the trunk parent interface (port #12)
<code>show version</code>	Confirms IOS version and uptime, useful if you suspect it crashed/reloaded

Fastest single check: `show running-config | include helper` — if this comes back empty, no DHCP relay is configured anywhere, which on its own would explain the symptom if your server and client are on different ports.

B. From any connected PC (works no matter what switch model you have)

OS	Command	What it tells you
Windows	<code>ipconfig /all</code>	Current IP, whether it's a real lease or 169.254.x.x self-assigned
Windows	<code>arp -a</code>	Devices the PC can see at Layer 2 — helps confirm what's reachable
Windows	<code>tracert <gateway-ip></code>	Whether traffic is routed (multiple hops) or local (one hop)
Linux/Mac	<code>ip addr / ip route</code>	Same as above for IP/gateway
Linux/Mac	<code>ip neigh (or arp -a)</code>	Same as Windows ARP table check

C. The switch-chip side (ports 1–12, separate from the Cisco router)

Be aware: this portion runs its own proprietary management software (referenced in the manual as a separate "Ethernet-Switch CLI user guide"), and the exact command syntax for that CLI isn't something I have reliable documentation for — guessing at exact syntax here could cost you time rather than save it. Two safer options:

- **Use the Web GUI instead** (192.168.1.111 per Section 2) — look for a VLAN membership/port table and a DHCP snooping/relay page; this is the most universal way to check port-to-VLAN assignment without needing exact CLI syntax.
- **If you do get into a CLI/serial prompt on the switch side**, type `?` or `help` at the prompt — nearly every network CLI supports this to list valid commands, and it'll give you the actual syntax for *that* firmware rather than a guess.

5. Quick diagnostic checklist

1. **Confirm it's really DHCP failing.** Check the device's address — `169.254.x.x` (Windows/most OSes) confirms DHCP timed out and the device self-assigned. No address at all, or an old static one, means something different is going on.
2. **Check which physical ports** the DHCP server and the failing client are plugged into.
3. **If on different ports** → almost certainly the VLAN/relay issue above.
4. **If on the same port/VLAN and still failing**, check for:
 - DHCP snooping enabled with the server's port marked "untrusted"
 - Spanning-tree port stuck in blocking/listening state (wait 30-60s after plugging in, or check port status)
 - Broadcast storm control throttling DHCP broadcasts
 - A port administratively shut down in the switch config
 - An ACL or firewall rule blocking UDP ports 67/68
5. **Hardware-level** (no config access needed):
 - Check the per-port Link/Activity LED — solid/flashing = physical link is up
 - Swap the cable and/or try a different port
 - Power-cycle in order: DHCP server device → switch → end devices, ~30 sec between each

6. Hard reset (last resort, 9012C/G only)

There's a physical reset function on Port #3:

- **Momentary** contact between Reset pin (M) and GND (L): graceful shutdown of the whole unit (switch + SBC). A second momentary contact powers it back up.
- **Holding for over 3 seconds**: forces a hard shutdown of switch and SBC. You'll need to disconnect and reconnect power afterward to bring it back up.

This does **not** reset configuration to factory defaults by itself — it's a power-cycle mechanism, not a config wipe. Use it if the unit is unresponsive, not as a way to clear settings.

Summary — fastest path under time pressure

1. Try static IPs (Section 0) to get connectivity working immediately, regardless of root cause.
2. If you have a few minutes, log into management (Section 2), and run `show running-config | include helper` and `show ip interface brief` (Section 4) to confirm the VLAN/relay theory (Section 3) in seconds.
3. If access fails or there's no time, keep DHCP server and critical clients on the same physical port/VLAN group as a structural workaround.